

ICHA, Russian Federation

WMO: 324110

Lat: 55.6137N

Lon: 155.6132E

Elev: 6

StdP: 101.26

Time Zone: 12.00 (E12)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.0	-22.0	-27.2	0.3	-23.6	-25.1	0.4	-21.8	15.0	-9.0	12.7	-4.9	3.5	30	0.838

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.8	17.0	15.1	15.7	14.3	14.7	13.6	15.6	16.3	14.6	15.3	13.7	14.5	4.3	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.2	10.8	15.9	14.1	10.1	14.9	13.4	9.6	14.3	43.9	16.3	40.9	15.5	38.7	14.6	22.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 13.1	11.4	10.0	DB	-26.5	20.8	2.3	2.5	-28.2	22.5	-29.5	24.0	-30.7	25.4	-32.4	27.2	(4)
(5)			WB	-26.4	18.2	2.2	2.2	-28.0	19.7	-29.3	21.0	-30.5	22.2	-32.1	23.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	0.4	-11.8	-10.5	-7.4	-1.9	2.9	7.6	11.5	12.2	9.0	4.1	-2.8	-8.9
	DBStd	9.20	5.83	5.76	4.35	3.10	2.24	2.32	2.10	2.24	2.35	2.95	4.40	5.41
	HDD10.0	3657	676	574	540	356	221	77	8	7	45	183	383	587
	HDD18.3	6549	934	807	798	606	478	322	213	190	281	441	633	846
	CDD10.0	149	0	0	0	0	0	6	53	76	14	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	1	0	0	0	0	0	0	0	1	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	4.7	4.3	4.4	4.5	4.6	4.8	4.1	3.9	4.5	4.6	5.1	6.1	5.1
Precipitation	PrecAvg	722	29	20	23	37	43	52	88	101	83	100	85	52
	PrecMax	1058	68	103	70	99	111	133	261	231	209	208	224	145
	PrecMin	433	3	1	3	4	5	0	6	10	14	29	22	4
	PrecStd	162	19	21	17	23	28	34	57	56	48	40	46	35
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.4	3.3	3.7	7.6	12.3	16.5	20.0	19.9	15.1	11.7	8.0	3.3
		MCWB	-0.2	0.4	0.8	4.4	7.9	11.7	17.2	17.7	13.0	8.1	6.5	0.4
	2%	DB	0.3	1.3	1.2	5.1	9.7	13.5	17.1	17.2	13.7	10.0	6.1	1.7
		MCWB	-1.7	-1.0	-0.6	2.4	6.5	10.9	15.4	15.4	12.2	8.2	4.8	-0.4
	5%	DB	-1.8	-0.5	-0.3	3.7	8.0	11.8	15.6	16.0	13.0	9.0	4.1	-0.3
		MCWB	-3.1	-2.3	-1.5	1.5	5.9	10.0	14.3	14.7	11.7	7.3	3.0	-1.6
	10%	DB	-3.8	-2.3	-1.5	2.4	6.5	10.7	14.5	15.1	12.4	8.2	2.9	-1.8
		MCWB	-4.6	-3.4	-2.4	0.7	4.9	9.4	13.5	14.1	11.2	6.9	1.8	-2.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.1	0.7	0.9	4.6	8.7	12.8	18.2	18.1	13.8	9.9	7.0	0.8
		MCDB	2.0	3.1	3.1	7.1	11.5	15.4	19.4	18.6	14.4	10.6	7.6	2.5
	2%	WB	-1.7	-0.9	-0.4	2.7	7.1	11.2	15.6	16.0	12.8	8.5	4.8	-0.4
		MCDB	-0.1	1.0	0.9	4.6	9.1	12.7	16.4	16.7	13.3	9.3	5.9	1.2
	5%	WB	-3.0	-2.1	-1.3	1.6	5.9	10.4	14.5	14.9	12.1	7.7	3.2	-1.3
		MCDB	-1.9	-0.7	-0.4	3.2	7.7	11.5	15.4	15.6	12.7	8.7	4.2	-0.2
	10%	WB	-4.6	-3.4	-2.3	0.9	4.9	9.6	13.6	14.1	11.5	7.0	1.9	-2.6
		MCDB	-3.7	-2.4	-1.6	2.2	6.3	10.6	14.4	14.8	12.1	8.1	2.7	-1.7
Mean Daily Temperature Range	5% DB	MDBR	5.8	6.1	6.0	5.1	4.5	3.9	3.8	3.8	4.7	4.3	4.4	5.4
		MCDBR	6.2	6.6	6.0	5.9	7.1	6.0	5.4	5.0	5.2	5.3	5.1	5.6
		MCWBR	5.5	5.2	4.9	4.1	4.5	4.0	4.0	3.5	3.9	4.2	4.4	4.8
	5% WB	MCDBR	5.8	6.4	5.7	5.2	6.2	5.4	4.9	4.2	4.0	4.1	4.8	5.3
		MCWBR	5.3	5.3	4.8	3.8	4.4	3.8	3.9	3.3	3.5	3.6	4.4	4.7
Clear-Sky Solar Irradiance	taub	0.272	0.311	0.357	0.433	0.476	0.445	0.460	0.432	0.367	0.363	0.327	0.278	
	taud	2.148	2.092	2.028	1.911	1.948	2.141	2.142	2.225	2.383	2.209	2.095	2.082	
	Ebn at Noon	631	736	786	773	766	801	778	776	787	667	530	507	
	Edh at Noon	59	93	127	166	170	142	140	121	90	83	64	50	
All-Sky Solar Radiation	RadAvg	0.60	1.31	2.69	3.86	4.74	4.84	4.46	3.40	2.74	1.47	0.64	0.40	
	RadStd	0.07	0.18	0.32	0.45	0.59	0.42	0.46	0.36	0.24	0.14	0.07	0.04	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page